

CS201 — Data Structures and Algorithms: Syllabus

Course Objectives

This course introduces the fundamental data structures and algorithmic techniques used to solve computational problems efficiently. Students learn to analyse the time and space complexity of algorithms, choose appropriate data structures, and implement them.

Module 1: Algorithm Analysis

Asymptotic notation: Big-O, Omega and Theta. Best, average and worst case analysis. Analysing loops and recurrences.

Module 2: Searching and Sorting

Linear search and binary search. Bubble sort, insertion sort, merge sort and quick sort with their complexities.

Module 3: Stacks and Queues

The stack ADT (LIFO) and the queue ADT (FIFO). Array and linked implementations. Applications: expression evaluation, recursion, undo operations, scheduling and breadth first search.

Module 4: Linked Lists

Singly, doubly and circular linked lists. Insertion, deletion, traversal and reversal.

Module 5: Trees

Binary trees, binary search trees and tree traversals. Balanced trees (AVL).

Module 6: Graphs

Graph representations, breadth first search, depth first search, shortest paths and topological ordering.

Module 7: Hashing

Hash functions, collisions, chaining and open addressing.

Assessment

Internal Assessment 1 (32 marks) covers Modules 2, 3, 4, 6 and 7. The end semester examination covers the full syllabus.